

SALFORD & CO.



SILICA



SILICA

FOUNDATIONAL TO COMPUTING, COMMUNICATIONS, MOBILITY, AND AI

SECURE & AUTOMOTIVE-GRADE SEMICONDUCTOR DESIGN
COMPANY
PUNE | INDIA

THE ENGINE OF MODERN TECHNOLOGY

PGPX 130 NITIN GHATKAR

BUSINESS MODELS IN THE INDUSTRY

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02

FABLESS: DESIGN ONLY

SMALLER NODES → HIGHER
PERFORMANCE & EFFICIENCY, BUT:

EXPONENTIALLY HIGHER
COST

MORE DESIGN
COMPLEXITY

LOWER MARGINS IF YIELDS ARE WEAK

Chipmaking requires extreme precision
and cleanroom conditions

"NODE" = PROCESS GENERATION
(E.G., 7NM, 5NM, 3NM)

01

IDM (INTEGRATED DEVICE
MANUFACTURER): DESIGN +
MANUFACTURE



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03

FOUNDRY:
MANUFACTURING ONLY

04

OSAT: OUTSOURCED
ASSEMBLY & TEST

THE PROBLEM & OPPORTUNITY



India's Semiconductor Design Opportunity

 **85-90%** Automotive Semiconductors Imported

 **20-40** Week Lead Times

MAIN CHIP CATEGORIES:



EV & Industrial Control Market

₹12,000 Cr+ Local Opportunity (3 Years)



 **₹76,000 Crore**
Govt. Semiconductor Mission

 **₹1.60 Lakh Crore**
Approved Investments

POLICY PUSH FOR INDIGENIZATION

Skilled Talent

Indigenization Push

Customized Solutions

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01 LOGIC (CPUS, GPUS)

02 MEMORY (DRAM, NAND)

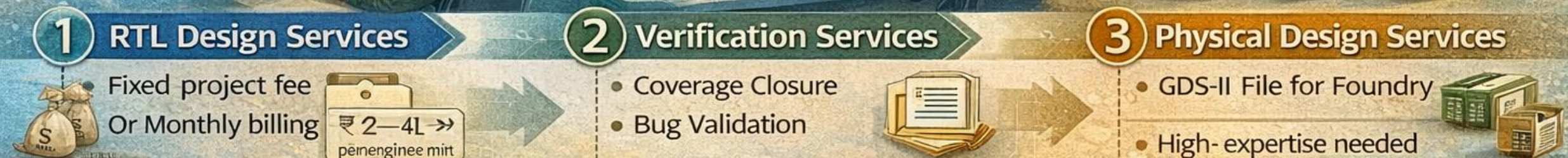
03 ANALOG & POWER (VOLTAGE REGULATION, SENSORS)



India has talent.
Customers exist.

But local secure
ASIC design
capacity is thin.

Services Business Flow for India's Fabless Semiconductor Venture



You deliver key steps, output to client's flow, and systematically gain IP know-how.

Builds revenue, then builds reusable IP.

DESIGN

EQUIPMENT & MATERIALS

MANUFACTURING

LITHOGRAPHY MACHINES,
CHEMICALS, WAFERS

ASSEMBLY, VERIFICATION,
ADVANCED PACKAGING

PACKAGING & TESTING

SEMICONDUCTOR VALUE CHAIN OVERVIEW

Market Definition

Target segment: EV + Industrial power/control semiconductors

Use cases: Motor control logic, BMS companion logic, charger controllers, power management control, Safety monitoring

Demand Drivers

EV growth (especially 2W/3W and charging infra)

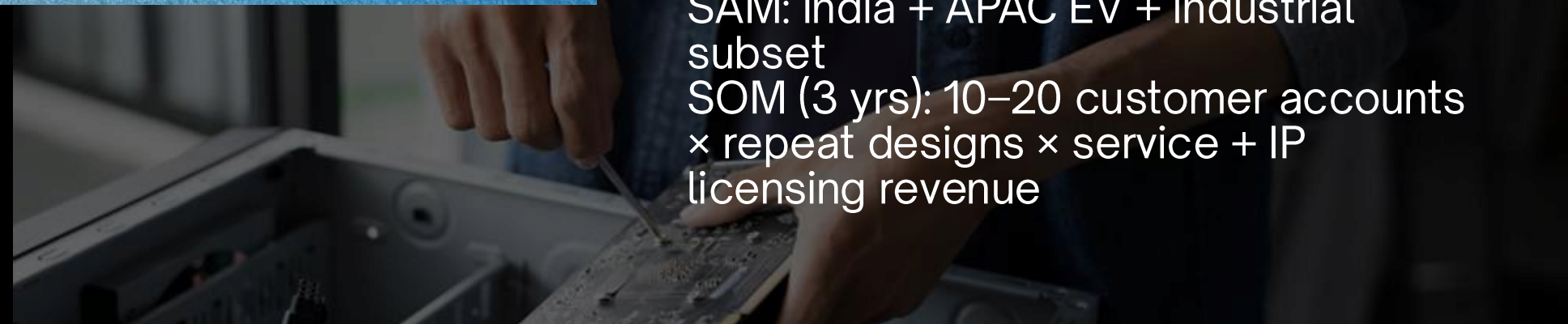
Industrial electrification: motors/drives/inverters everywhere
Cost and supply risk pushes localization

TAM-SAM-SOM

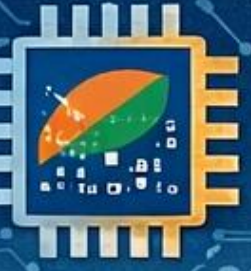
TAM: Global power + Control Semis market

SAM: India + APAC EV + industrial subset

SOM (3 yrs): 10-20 customer accounts
× repeat designs × service + IP licensing revenue



WHY SEMICONDUCTORS MATTER



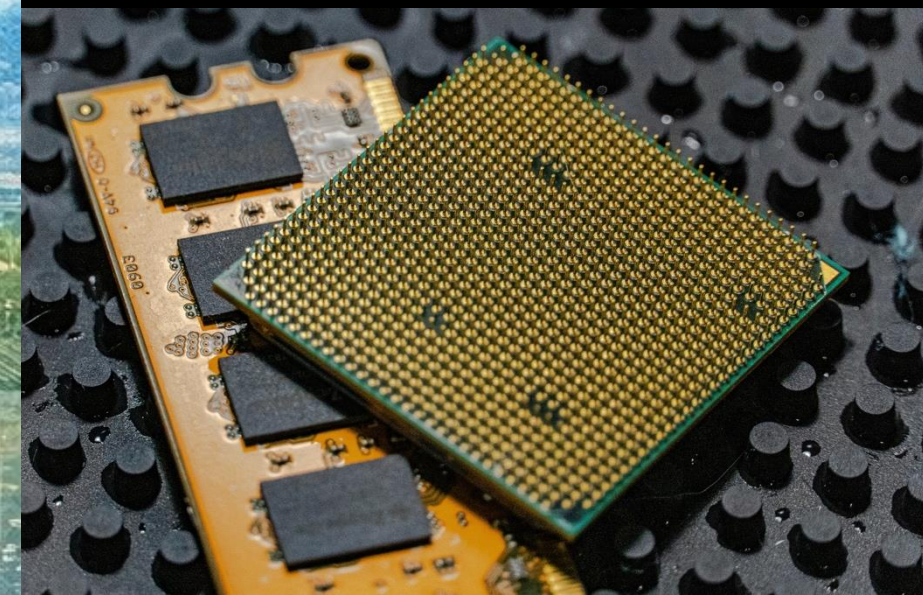
**INDIA'S
SEMICONDUCTOR
MISSION**

Competitor Benchmarking: Design Services (Year 1-2) India's Fabless Custom Semiconductor Design

Global giants win scale; service firms win manpower. We win India-specific customization + reusable IP accumulation.

▶ STRONG LINK TO
GDP GROWTH
AND INNOVATION

Global Product Giants	India Design Services		Indian Fabless Precedents	INDIA'S Fabless Custom
Infineon ST TEXAS INSTRUMENTS RENESAS onsemi	bos wipro TATA ELXSI	TCS wipro TATA ELXSI	INDIA'S Fabless Custom	
Speed + Customization	★★★★	★★★★	★★★★★★★★	★★★★
Local Appt Engineering	★★★★	★★★★	★★★★★★★★	★★★★
Qualification Readiness	★★★★	★★★★	General	EV Power / Control
Domain Focus	Volume	Manpower	★★★★	★★
Pricing	Volume	Product		OEM Access
Sales Motion	Tier-1/OEM	Tier-1/OEM	Semi/Startup Ecosystem	★★★★



Foundry/Tapeout Readiness: ★★★★★  Global giants win scale; service firms win manpower. We win India-specific customization + reusable IP accumulation.

TEAM SIZE

India has talent.
Customers exist.

- ▶ But local secure ASIC design capacity is thin.

MAIN CHIP CATEGORIES:

POLICY PUSH FOR
INDIGENIZATION



Phase-Wise Team Size

PHASE 1: SERVICES FOCUS

LEAN SURVIVAL MODE

- CTO / Arch Lead
- 2 RTL
- 2 Verification
- 1 Physical Design



6–8 HIGH-SKILL TEAM

- Engineering Core
- Add engineers only after:
 - Signed contracts
 - 50%+ utilization confirmed

02

MEMORY (DRAM, NAND)

PHASE 2: IP

03

ANALOG & POWER (VOLTAGE REGULATION, SENSORS)

YEAR 2

SCALABLE IP DEVELOPMENT

- CTO
- 3 RTL
- 4 Verification
- 3 Physical Design
- + Verification Lead
- + IP Docs Owner
- + Process Engineer

SCALE TO 12–15 TEAM

- Add engineers only after:
- Signed contracts
- 50%+ utilization confirmed

RULE OF THUMB: Start lean with 6–8 member core team – signed contracts + confirmed 50%+ utilization.



INDIA'S FABLESS SEMICONDUCTOR VENTURE

Business Model

Phase 1: Design Services (Revenue First)

- RTL design services
- Verification (UVM) services
- Physical design services

Pricing:

- Per-project fixed fee
- Monthly Billing
₹ engineer-month

Phase 2: Reusable IP Blocks (Own Assets)

- Reuse logic across customers
- License blocks (license fee + optional royalty) + (optional royalty)

Phase 3: India-Optimized Product (Fabless)

- Niche products for mid-volume OEMs
- Sell through distributors + OEM design wins



INDIA'S FABLESS CUSTOM SEMICONDUCTOR VENTURE

Early survival requires shared/limited tool access or an anchor client.

Break-Even Analysis

Low-Tool vs High-Tool

Assumptions:

- Team: 6 Engineers, 5 Billable
- Billing Rate: ₹ 3.5L per billable engineer/month
- Utilization Ramp: 20% (M1-3), 50% (M4-6), 70%



Revenue Per Month

- M1-3: ₹ 3.5L
- M4-6: ₹ 8.75L
- M7-12: ₹ 12.25L

Months

Low-Tool Burn - ₹ 17L

- Low-Tool Model - ₹ 17L/month
- High-Tool Model - ₹ 26L/month

Burn per month: 97% High-Tool

Full tool stack
upfront makes
break-even
mathematically
unrealistic.

THE NEXT WAVE OF
TECHNOLOGY
DEPENDS ON THE
NEXT WAVE OF CHIPS.

THE NEXT WAVE OF TECHNOLOGY DEPENDS ON THE NEXT WAVE OF CHIPS.



INDIA'S SEMICONDUCTOR MISSION

Analysis: India's Fabless Semiconductor Design

1 Phase 1: Design Services (Revenue-First)

- RTL Design
- Verification (UVM)
- Physical Design

2 Phase 2: Reusable IP Blocks (Own assets)

- Reuse logic across customers
- License blocks (license fee + optional royalty)

3 Phase 3: India-Optimized Products (Fabless)

- Niche products for mid-volume OEMs
- Sell through distributors + OEM design wins



Break-Even Analysis: Low-Tool vs High-Tool

Revenue per Month

(5 Billable Engineers, Gradual Utilization Ramp)

M1~3	M4~6	M7~12	Burn per
₹3.5 L	₹8.75 L	₹12.25 L	Billable x Rate x Utiliz.

Billable × Rate × Utilization

Burn per Month

Low-tool model
(shared/limited tools)
conservative burn)

$$\text{₹ } 17 \text{ L} / 5 \times 5 \times 3.5 \text{ L} = \sim 97\%$$

Break-even util.

Low-tool model
(full EDA licenses)
high burn)

$$\text{₹ } 26 \text{ L} / 5 \times 5 \times 3.5 \text{ L} = \sim 149\%$$

Impossible

THANK YOU

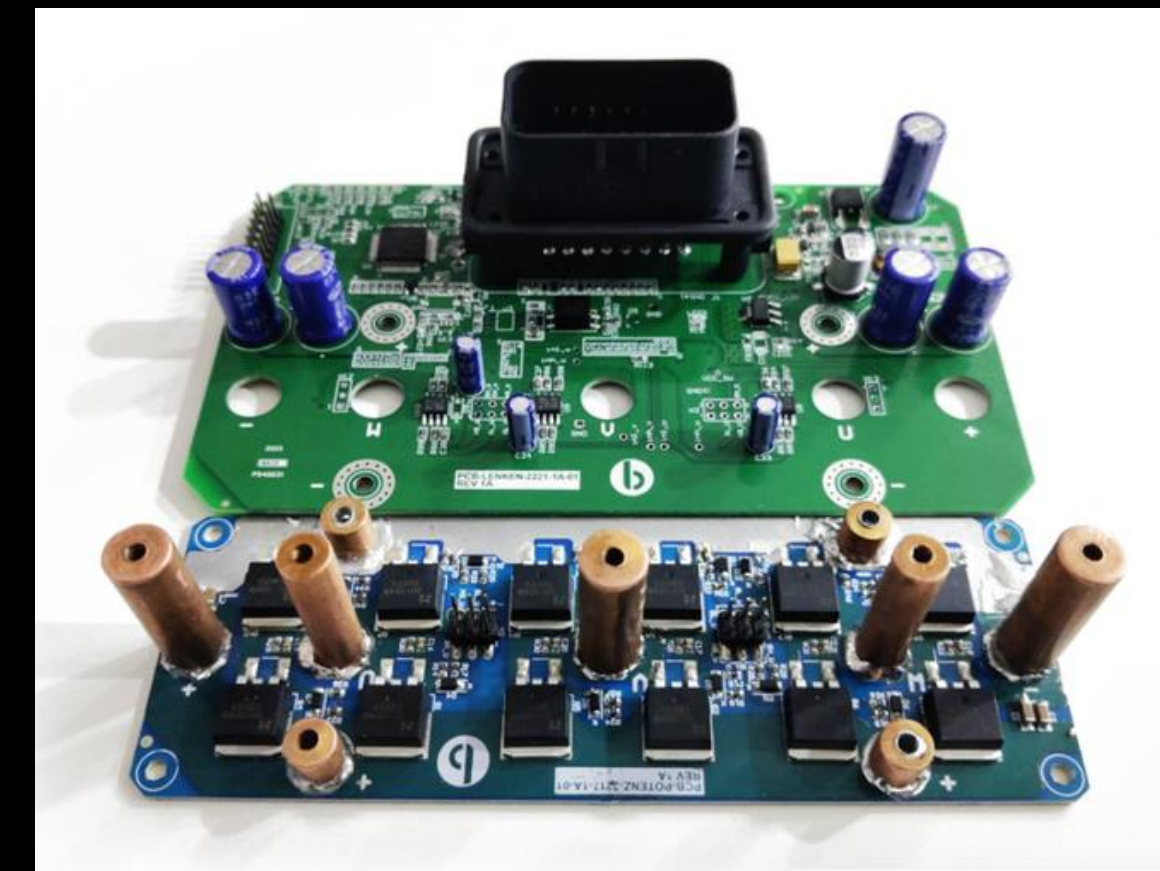
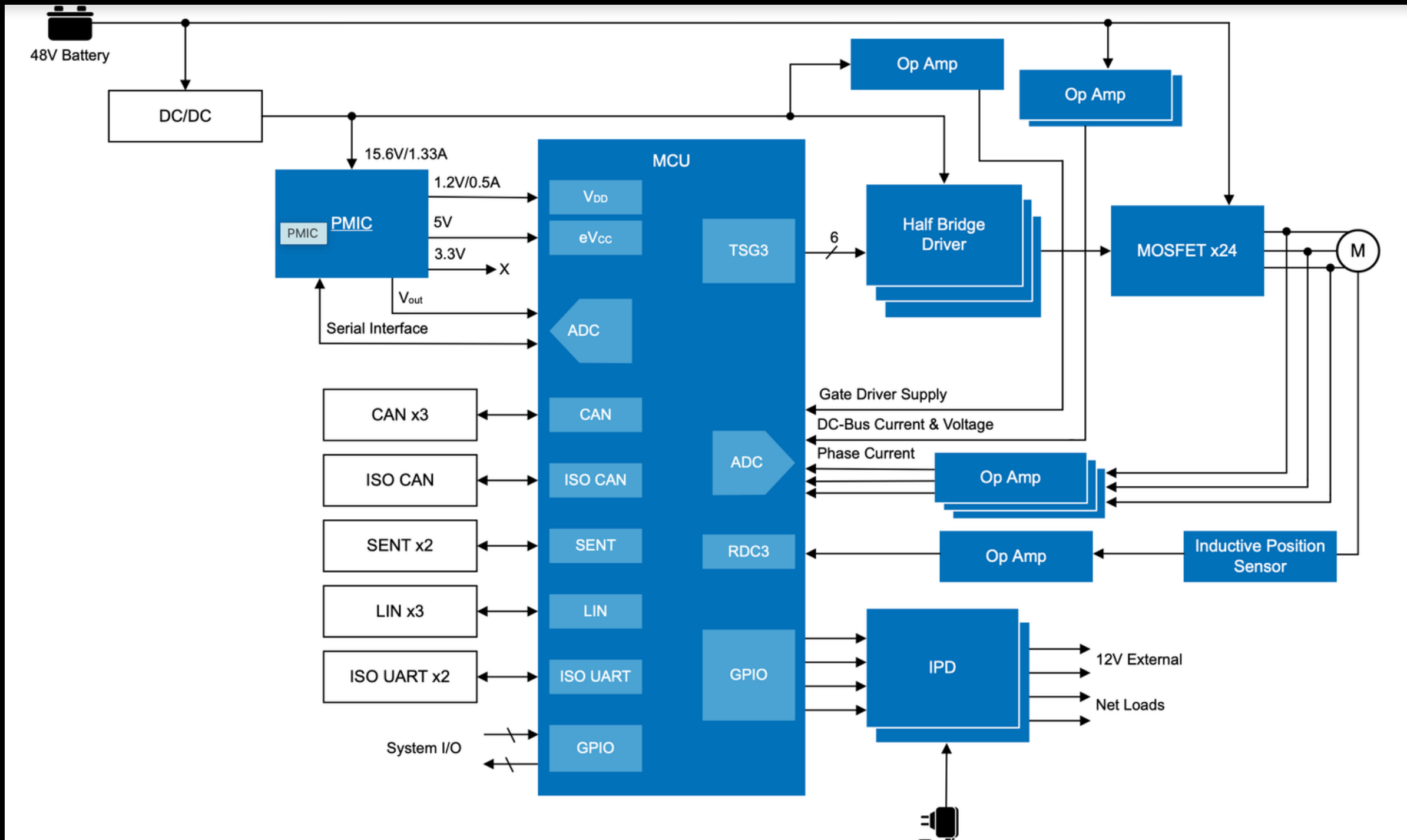
01

02

FABLESS: DESIGN ONLY

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04



MO T OR
CONTROLLER